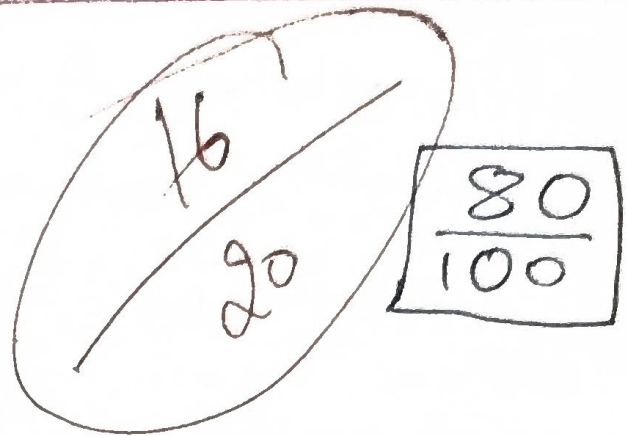


\\CSA-1704\\ \\Artificial Intelligence\\



1. Smart home automation system:

⇒ Smart home automation system is a Ai based system which makes the work easy and helps to save time current.

⇒ Artificial intelligence implimented on many advanced machines, help for ~~smart~~ smart home automation system:

(i) Simple reflex:

⇒ It is on basic automation system: which should be implimented in every smart home.

⇒ It is on automatic sensing device which helps to turn on or turn off the light when any human being is sensed by motion.

⇒ When someone enters the room the light automatically turns on by sensing them by using motion sensor, and it turns off after the motion is off.

Advantages:-

- * It helps to save electricity
- * It makes the work easier.
- * It also helps to reduce pollutions induced

Disadvantages:-

⇒ Sometimes if sensor gets repair of a motion is often sensed the light will be in turn on forever.

⇒ It may increase the electricity usage.

Other Implementations:-

⇒ There are many AI implementations like cleaning, AI assistant, weather predicting agent for helps for agriculture and even in smart home for solar generator heaters.

⇒ All AI implemented tools and devices helps for reducing human errors and built on smart home automation system.

A Robot is Placed in Unknown Mazo :-

→ Robots are futuristic machines that helps humans in many ways but a very intelligence robot named Sofia. after implimenting all the advanced informations, after defeating the human in a chess game, the first word it said as, the time is started for robots to completely destroy the human industry.

- (i) Uniform Cost Search [UCS] :-
- (ii) Iterative Deeping Search [IDS] :-

These are the two problem solving methods

Hence there are many problem solving methods, we choose these both methods because it is depending on important algorithms and so' also in space complexity.

(i) Uniform Cost Search (UCS):-

- * Expands the node with the lowest path cost first, * uses a priority queue.
- * Guarantees the least-cost path.

Advantage:

- * Complete algorithm.
- * Always finds the optimal solution.

Disadvantages:

- * High memory usage.
- * Slow for very large search spaces.

Complexity

- Time complexity : $O[b^{(1+LC^*/e)}]$
- Space complexity : $O[b^{(1+LC^*/e)}]$

Iterative Deeping Search (IDS) :-

- Performs Depth-Limited search repeatedly.
- Increases depth limit until the goal.

Advantage:-

- Requires very little memory.
- Suitable when the goal depth is unknown.

Complexity

- Time complexity : $O(b^d)$
- Space Complexity : $O(bd)$